
Retail Sales Performance Analysis Report

An End-to-End Data Analytics Project
Using SQL, Python, Excel and Power BI

Dataset Superstore Sales Dataset (Kaggle)
Records 9,994 transactions (2014–2017)
Tools SQL · Python · Excel · Power BI
Domain Retail / E-commerce Analytics
Author Abhishek Kumar
Date June 24, 2026

Key Results at a Glance

\$2.30M Revenue

\$286K Profit

9,994 Orders

12.47% Margin

Contents

1	Introduction	3
1.1	Background	3
1.2	Problem Statement	3
1.3	Objectives	3
2	Dataset Description	3
2.1	Source	3
2.2	Features Overview	4
3	Tools & Technologies	4
4	Project Workflow	4
5	Phase 1: SQL Analysis	5
5.1	Overview	5
5.2	Query 1: Business Overview	5
5.3	Query 2: Sales & Profit by Region	5
5.4	Query 3: Sales by Product Category	6
5.5	Query 4: Impact of Discount on Profit	6
5.6	Summary of All SQL Queries	7
6	Phase 2: Python EDA	7
6.1	Environment	7
6.2	Libraries Used	7
6.3	Data Loading & Cleaning	7
6.4	Data Quality Checks	8
6.5	Charts Generated	8
7	Phase 3: Excel Report	8
7.1	Overview	8
7.2	Pivot Tables Created	8
7.3	Key Excel Features Used	8
8	Phase 4: Power BI Dashboard	9
8.1	Overview	9
8.2	DAX Measures Created	9
8.3	Visuals in Dashboard	9
9	Key Business Insights	9
9.1	Revenue & Profit	9
9.2	Regional Performance	10
9.3	Product Category Insights	10
9.4	Discount Impact	10
9.5	Customer Segments	10
9.6	Shipping Analysis	10
9.7	Sales Trend	10

10 Recommendations	11
11 Project Structure	11
12 How to Run This Project	12
12.1 SQL	12
12.2 Python (Google Colab)	12
12.3 Excel	12
12.4 Power BI	12
13 Conclusion	12

1 Introduction

1.1 Background

In today's competitive retail landscape, data-driven decision-making is critical for sustaining profitability and growth. This project performs a comprehensive analysis of a retail superstore's sales data spanning four years (2014–2017), covering 9,994 transactions across the United States.

The goal is to simulate a real-world data analyst workflow — from raw data ingestion and SQL querying, through Python-based exploratory analysis, to Excel reporting and interactive Power BI dashboards.

1.2 Problem Statement

The retail business faces several unresolved questions:

- Which regions and product categories drive the most revenue and profit?
- Are discount strategies helping or hurting overall profitability?
- Which products are consistently unprofitable and should be reviewed?
- What are the seasonal sales trends across years?
- Which customer segments represent the highest value?
- How do different shipping modes affect delivery time and customer experience?

1.3 Objectives

1. Extract and analyse sales data using **SQL** queries
2. Perform data cleaning and exploratory analysis using **Python**
3. Build summary reports and pivot tables in **Excel**
4. Create an interactive business dashboard in **Power BI**
5. Derive actionable business insights and recommendations

2 Dataset Description

2.1 Source

The dataset used is the **Superstore Sales Dataset** publicly available on Kaggle. It represents sales transactions of a fictional US-based retail superstore.

Attribute	Details
Source	Kaggle — Superstore Sales Dataset
Records	9,994 rows
Columns	21 features
Time Period	January 2014 – December 2017
Geography	United States (4 regions, 49 states)
File Format	CSV / Excel (.xlsx)

2.2 Features Overview

Column	Type	Description
Order ID	Text	Unique order identifier
Order Date	Date	Date the order was placed
Ship Date	Date	Date the order was shipped
Ship Mode	Text	Shipping method selected
Customer ID	Text	Unique customer identifier
Customer Name	Text	Full name of customer
Segment	Text	Consumer / Corporate / Home Office
Region	Text	East / West / Central / South
Category	Text	Furniture / Office Supplies / Technology
Sub-Category	Text	Product sub-group (17 types)
Product Name	Text	Name of the product sold
Sales	Numeric	Revenue generated (\$)
Quantity	Numeric	Units ordered
Discount	Numeric	Discount applied (0 to 1)
Profit	Numeric	Net profit (\$)

3 Tools & Technologies

Tool	Version	Purpose
SQLite / DB Browser	3.x	Data storage and SQL querying
Python	3.10+	Data cleaning and EDA
pandas	2.x	Data manipulation
matplotlib	3.x	Data visualisation
seaborn	0.x	Statistical plots
Google Colab	–	Notebook environment
Microsoft Excel	2021	Pivot tables and reporting
Power BI Desktop	Latest	Interactive dashboard
GitHub	–	Version control and portfolio

4 Project Workflow

The project follows a standard end-to-end data analytics pipeline:

Raw CSV Data → SQL Analysis → Python EDA → Excel Report → Power BI

5 Phase 1: SQL Analysis

5.1 Overview

The dataset was imported into **DB Browser for SQLite** and queried using 10 analytical SQL queries. Each query was designed to answer a specific business question.

5.2 Query 1: Business Overview

Listing 1: Business Overview

```
1 SELECT
2     COUNT(DISTINCT "Order ID") AS total_orders,
3     ROUND(SUM(Sales), 2)       AS total_revenue,
4     ROUND(SUM(Profit), 2)      AS total_profit,
5     ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS profit_margin_pct
6 FROM "Sample - Superstore";
```

Result:

Total Orders	Total Revenue	Total Profit	Profit Margin
5,009	\$2,297,200.86	\$286,397.02	12.47%

5.3 Query 2: Sales & Profit by Region

Listing 2: Sales and Profit by Region

```
1 SELECT
2     Region,
3     COUNT(DISTINCT "Order ID") AS total_orders,
4     ROUND(SUM(Sales), 2)       AS revenue,
5     ROUND(SUM(Profit), 2)      AS profit,
6     ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS margin_pct
7 FROM "Sample - Superstore"
8 GROUP BY Region
9 ORDER BY revenue DESC;
```

Result:

Region	Orders	Revenue	Profit	Margin %
West	1,611	\$725,457	\$108,418	14.94%
East	1,401	\$678,781	\$91,522	13.48%
Central	1,175	\$501,239	\$39,706	7.92%
South	822	\$391,721	\$46,749	11.93%

5.4 Query 3: Sales by Product Category

Listing 3: Sales by Product Category

```

1 SELECT
2     Category,
3     COUNT(DISTINCT "Order ID") AS total_orders,
4     SUM(Quantity)              AS units_sold,
5     ROUND(SUM(Sales), 2)       AS revenue,
6     ROUND(SUM(Profit), 2)      AS profit,
7     ROUND(SUM(Profit) / SUM(Sales) * 100, 2) AS margin_pct
8 FROM "Sample - Superstore"
9 GROUP BY Category
10 ORDER BY revenue DESC;

```

Result:

Category	Units Sold	Revenue	Profit	Margin %
Technology	6,939	\$836,154	\$145,454	17.40%
Furniture	8,028	\$741,999	\$18,451	2.49%
Office Supplies	22,906	\$719,047	\$122,490	17.03%

5.5 Query 4: Impact of Discount on Profit

Listing 4: Discount Impact Analysis

```

1 SELECT
2     CASE
3         WHEN Discount = 0      THEN '0% - No Discount'
4         WHEN Discount <= 0.1   THEN '1-10%'
5         WHEN Discount <= 0.2   THEN '11-20%'
6         WHEN Discount <= 0.3   THEN '21-30%'
7         ELSE '31%+'
8     END AS discount_band,
9     COUNT(*)              AS total_items,
10    ROUND(AVG(Profit), 2) AS avg_profit,
11    ROUND(SUM(Profit), 2) AS total_profit
12 FROM "Sample - Superstore"
13 GROUP BY discount_band
14 ORDER BY discount_band;

```

Key Finding: Orders with discounts above 20% become loss-making on average.

5.6 Summary of All SQL Queries

#	Query Name	Business Question Answered
1	Business Overview	What are the overall KPIs?
2	Sales by Region	Which region performs best?
3	Sales by Category	Which category is most profitable?
4	Monthly Sales Trend	What are the seasonal patterns?
5	Top 10 Products by Revenue	What are the best-selling products?
6	Least Profitable Products	Which products lose money?
7	Customer Segment	Which segment is most valuable?
8	Shipping Days Analysis	How fast are orders shipped?
9	Discount Impact	Do discounts hurt profitability?
10	Top 10 High-Value Customers	Who are the best customers?

6 Phase 2: Python EDA

6.1 Environment

All Python analysis was performed in **Google Colab** using a Jupyter Notebook (`EDA.ipynb`).

6.2 Libraries Used

Listing 5: Library Imports

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4 import warnings
5 warnings.filterwarnings('ignore')
```

6.3 Data Loading & Cleaning

Listing 6: Data Loading and Feature Engineering

```
1 # Load data
2 df = pd.read_csv('Sample - Superstore.csv',
3                 encoding='latin1',
4                 parse_dates=['Order Date', 'Ship Date'])
5
6 # Feature Engineering
7 df['Year'] = df['Order Date'].dt.year
8 df['Month'] = df['Order Date'].dt.to_period('M').astype(str)
9 df['Ship Days'] = (df['Ship Date'] - df['Order Date']).dt.days
10 df['Discount Band'] = pd.cut(df['Discount'],
11                             bins=[-0.01, 0, 0.1, 0.2, 0.3, 1.0],
12                             labels=['0%', '1-10%', '11-20%', '21-30%', '31%+'])
```

6.4 Data Quality Checks

Check	Result
Missing Values	0 (None)
Duplicate Rows	0 (None)
Total Records	9,994
Date Range	Jan 2014 – Dec 2017
Unique Customers	793
Unique Products	1,850

6.5 Charts Generated

1. **Chart 1 — KPI Overview Cards:** Total Revenue, Profit, Orders, Margin displayed as metric cards
2. **Chart 2 — Revenue by Region:** Grouped bar and pie chart showing regional performance
3. **Chart 3 — Sales by Category:** Horizontal bar for categories, vertical bar for top sub-categories
4. **Chart 4 — Monthly Trend:** Dual-axis line chart showing revenue and profit over time
5. **Chart 5 — Discount Impact:** Scatter plot and grouped bar showing discount vs profit relationship
6. **Chart 6 — Customer Segments:** Grouped bar and order count by segment

7 Phase 3: Excel Report

7.1 Overview

The dataset was loaded into Microsoft Excel and analysed using **Pivot Tables** and **Pivot Charts**.

7.2 Pivot Tables Created

Sheet	Pivot Table	Fields Used
PT - Region	Sales by Region	Region, Sales, Profit, Quantity
PT - Category	Sales by Category	Category, Sub-Category, Sales, Profit
PT - Monthly	Monthly Trend	Order Date (grouped), Sales, Profit
PT - Segment	Customer Segments	Segment, Sales, Profit, Orders

7.3 Key Excel Features Used

- Format as Table (Ctrl+T) for dynamic ranges

- Calculated field for Profit Margin % in pivot tables
- PivotCharts — Clustered Bar and Pie charts
- Currency number formatting on all financial columns
- Dashboard sheet with all 4 charts arranged in a 2×2 grid

8 Phase 4: Power BI Dashboard

8.1 Overview

An interactive business dashboard was built in **Power BI Desktop** using the Superstore dataset, featuring 7 visuals and 5 KPI cards.

8.2 DAX Measures Created

Listing 7: DAX Measures in Power BI

```
1 Total Revenue = SUM('Sample - Superstore'[Sales])
2 Total Profit  = SUM('Sample - Superstore'[Profit])
3 Total Orders  = DISTINCTCOUNT('Sample - Superstore'[Order ID])
4 Total Customers = DISTINCTCOUNT('Sample - Superstore'[Customer ID])
5 Profit Margin % =
6     DIVIDE([Total Profit], [Total Revenue]) * 100
```

8.3 Visuals in Dashboard

#	Visual Type	Content
1	Card (×5)	Revenue, Profit, Orders, Customers, Margin %
2	Clustered Bar	Total Revenue & Profit by Region
3	Donut Chart	Total Revenue by Category
4	Line Chart	Revenue & Profit trend by Year
5	Pie Chart	Revenue by Customer Segment
6	Table	Product-level Order Date and Sales
7	Slicer	Filter by Order Date / Year

9 Key Business Insights

9.1 Revenue & Profit

Total Revenue: \$2,297,200

Total Profit: \$286,397

Margin: 12.47%

9.2 Regional Performance

- **West** region leads in both revenue (\$725K) and profit margin (14.94%)
- **Central** region has the lowest margin at just 7.92% — needs immediate attention
- **South** region has fewer orders (822) but a healthy 11.93% margin

9.3 Product Category Insights

- **Technology** is the most profitable category (\$145K profit, 17.4% margin)
- **Furniture** generates high revenue (\$741K) but very low profit (\$18K — only 2.49% margin)
- **Tables** sub-category is loss-making at **−\$17,725** total profit
- **Bookcases** also unprofitable at **−\$3,472**

9.4 Discount Impact

- Orders with **0% discount** generate the highest average profit
- Once discounts exceed **20%**, average profit turns **negative**
- Heavy discounting (31%+) results in significant losses
- **Recommendation:** Cap discounts at 20% maximum

9.5 Customer Segments

- **Consumer** segment dominates with 50.56% of total revenue (\$1.16M)
- **Corporate** contributes 30.74% (\$706K)
- **Home Office** is smallest at 18.7% (\$430K) but often has higher order values

9.6 Shipping Analysis

- **Same Day** shipping is fastest but least used
- **Standard Class** is most popular, averaging 5 days delivery
- Most customers accept Standard Class — opportunity to upsell faster shipping

9.7 Sales Trend

- Revenue shows consistent **year-over-year growth** from 2014 to 2017
- **Q4** (October–December) is consistently the strongest sales period
- **Q1** (January–February) is consistently the weakest — good period for promotions

10 Recommendations

1. **Review Furniture Pricing:** Tables and Bookcases are losing money despite high sales volume. Renegotiate supplier costs or revise pricing strategy.
2. **Cap Discounts at 20%:** Data clearly shows discounts above 20% destroy profit margins. Implement a discount approval policy.
3. **Invest in the West Region:** West has the best margin (14.94%). Allocate more marketing budget to amplify growth here.
4. **Improve Central Region Profitability:** Central has the worst margin (7.92%). Investigate high costs or aggressive discounting in this region.
5. **Focus on Technology Products:** Technology delivers the best margins. Expand product range and promote high-margin tech items.
6. **Q1 Promotions Strategy:** Sales drop significantly in January–February. Run targeted campaigns during this slow period to maintain revenue consistency.
7. **Corporate Segment Upsell:** Corporate customers place fewer but larger orders. Create dedicated account management and loyalty programs.

11 Project Structure

```
retail-sales-analytics/  
|  
|-- SQL/  
|   |-- 01_Business_Overview.sql  
|   |-- 02_Sales_Profit_by_Region.sql  
|   |-- 03_Sales_by_Product_Category.sql  
|   |-- 04_Monthly_Sales_Trend.sql  
|   |-- 05_Top_10_Products_by_Revenue.sql  
|   |-- 06_Least_Profitable_Products.sql  
|   |-- 07_Customer_Segment_Performance.sql  
|   |-- 08_Average_Shipping_Days.sql  
|   |-- 09_Impact_of_Discount_on_Profit.sql  
|   +-- 10_Top_10_High_Value_Customers.sql  
|  
|-- Python/  
|   +-- EDA.ipynb  
|  
|-- Excel/  
|   +-- Sample_Superstore.xlsx  
|  
|-- PowerBI/  
|   +-- Retail_Sales_Dashboard.pbix  
|  
+-- README.md
```

12 How to Run This Project

12.1 SQL

1. Download **DB Browser for SQLite** from sqlitebrowser.org
2. Import `Sample - Superstore.csv` via File → Import → Table from CSV
3. Open the `Execute SQL` tab
4. Copy-paste any `.sql` file and press F5 to run

12.2 Python (Google Colab)

1. Open colab.research.google.com
2. Upload `EDA.ipynb`
3. Upload `Sample - Superstore.csv` when prompted
4. Run all cells with Runtime → Run All

12.3 Excel

1. Open `Sample.Superstore.xlsx` in Microsoft Excel
2. Navigate to each PT - sheet to view Pivot Tables
3. Open the Dashboard sheet for the overview

12.4 Power BI

1. Install **Power BI Desktop** (free) from Microsoft
2. Open `Retail_Sales_Dashboard.pbix`
3. Use the Year slicer to filter data interactively

13 Conclusion

This project demonstrates a complete end-to-end data analytics workflow applied to real-world retail sales data. Through SQL querying, Python exploratory analysis, Excel pivot reporting, and Power BI dashboarding, we extracted meaningful business insights from 9,994 transactions spanning four years.

The most critical findings include the negative impact of heavy discounting on profitability, the underperformance of the Furniture category despite high sales volumes, and the consistent growth trend in the West region. These insights translate directly into actionable business recommendations that a data analyst would present to stakeholders.

Skills Demonstrated in This Project

SQL Querying	Python (pandas)	Data Visualisation	DAX Measures
Pivot Tables	EDA	Business Reporting	Dashboard Design

End of Report · [Your Name] · June 24, 2026